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Palaeomagnetic Study of Dated Pre-Cambrian Rocks of the Front Range, Colorado–Wyoming

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Summary

The mean direction of remanent magnetization, after partial a.c. demagnetization, of Pre-Cambrian rocks from the Front Range of Colorado and Wyoming gives a palaeomagnetic pole position at 8 °N, 29 °E. Rock types sampled include andesite dykes, diorite inclusions in granite, and a highly oxidized red granite, all with a radiogenic age of 1·41 G.Y. This preliminary study suggests there are at least two and probably as many as four field reversals recorded in these rocks, and that remanent directions from andesite dykes may be useful in determining tectonic rotation of the Front Range.

1. Introduction

Palaeomagnetic data for Pre-Cambrian rocks in North America are available from the south-west United States (Runcorn 1964, DuBois 1964), Montana (in Irving 1964), Missouri (Hays & Scharon 1966), and several parts of Canada (Fahrig *et al.* 1965). However, palaeomagnetic directions are singularly lacking from the widespread Pre-Cambrian exposures in the Rocky Mountain region of Colorado and Wyoming. The paucity of data can be attributed principally to the types of rocks exposed; the almost entirely high-grade metamorphic and granitic rocks are not particularly suitable for palaeomagnetic study. The magnetization of metamorphic rocks is poorly understood; the magnetization of granites generally resides in large, homogeneous grains, which are magnetically unstable.

However, in the northern Front Range, Colorado–Wyoming, stable, well-grouped directions have been obtained from three different rock types: red granite, andesite dykes, and fine-grained xenoliths in granitic rocks.

2. General geology

The Front Range, which is the eastern-most swell of the Colorado Rocky Mountains, rises sharply from the interior plains to a maximum height of 14 255 ft (see Fig. 1). The range branches as it reaches into Wyoming, where the eastern prong is called the Laramie Range.

Most of the samples in this study were collected near the Colorado–Wyoming state line in the northern Front Range (Fig. 2). The range consists largely of folded metamorphic rocks intruded by two granitic batholiths, the Sherman and Log

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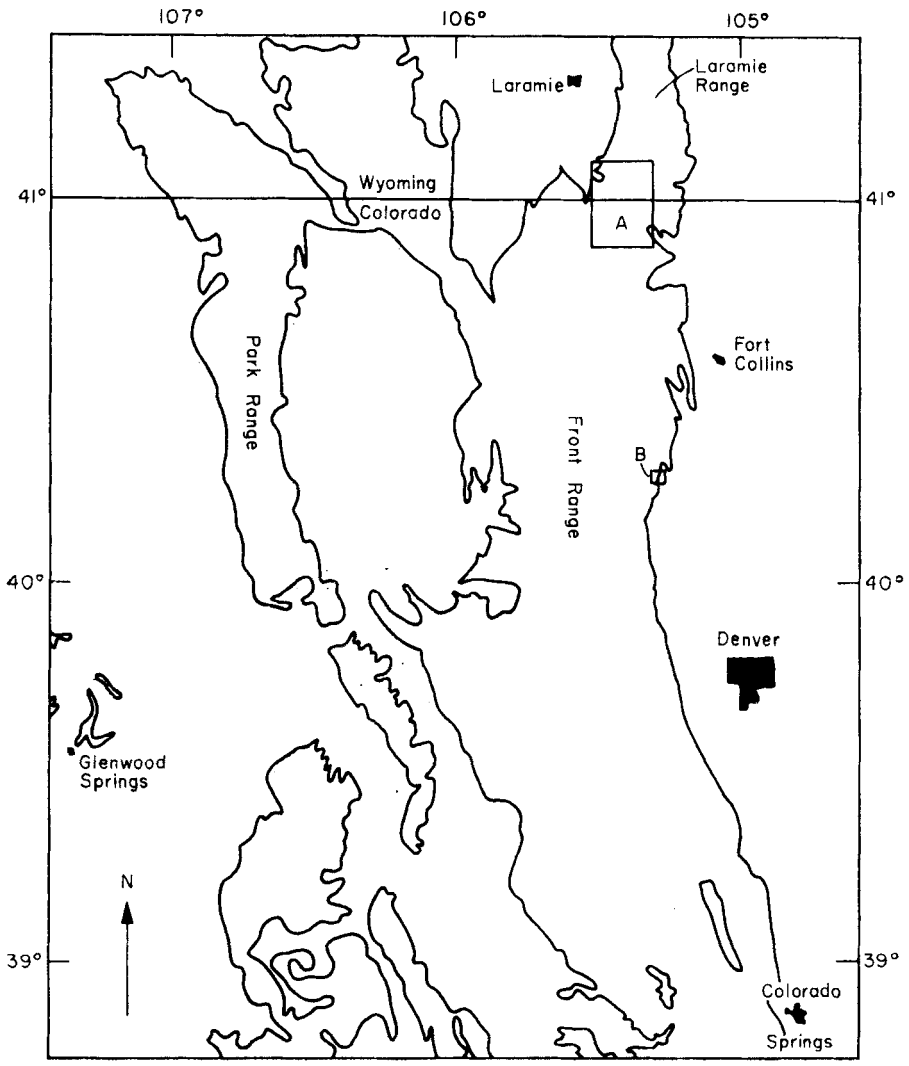


FIG. 1. Location of map of areas studied in north-central Colorado and southern Wyoming. Outcrops of Pre-Cambrian rocks are outlined. Sub-areas A and B are shown in Figs 2 and 3.

Cabin. The Sherman Granite, the older of the two, is of principal interest in this study. It is normally red, equigranular granite and forms the bulk of the Sherman batholith in the southern Laramie Range. Near Virginia Dale, Colorado, Sherman Granite was intruded in three circular zones, which form a ring-dyke complex about nine miles across (Eggler 1967). The outer zone or dyke is similar to the main Sherman batholith. The core of the complex, which is separated from the outer dyke by a screen of brecciated diorite, consists of inner and outer zones of porphyritic quartz monzonite, which is relatively younger than the red granite. The outer zone, which is slightly finer grained than the inner, may be the older of the two.

Sherman Granite is cut by a swarm of north-west-trending andesite dykes. The swarm continues south to the Big Thompson canyon area, where the dykes are cut by a granite correlated with that of the Log Cabin batholith.

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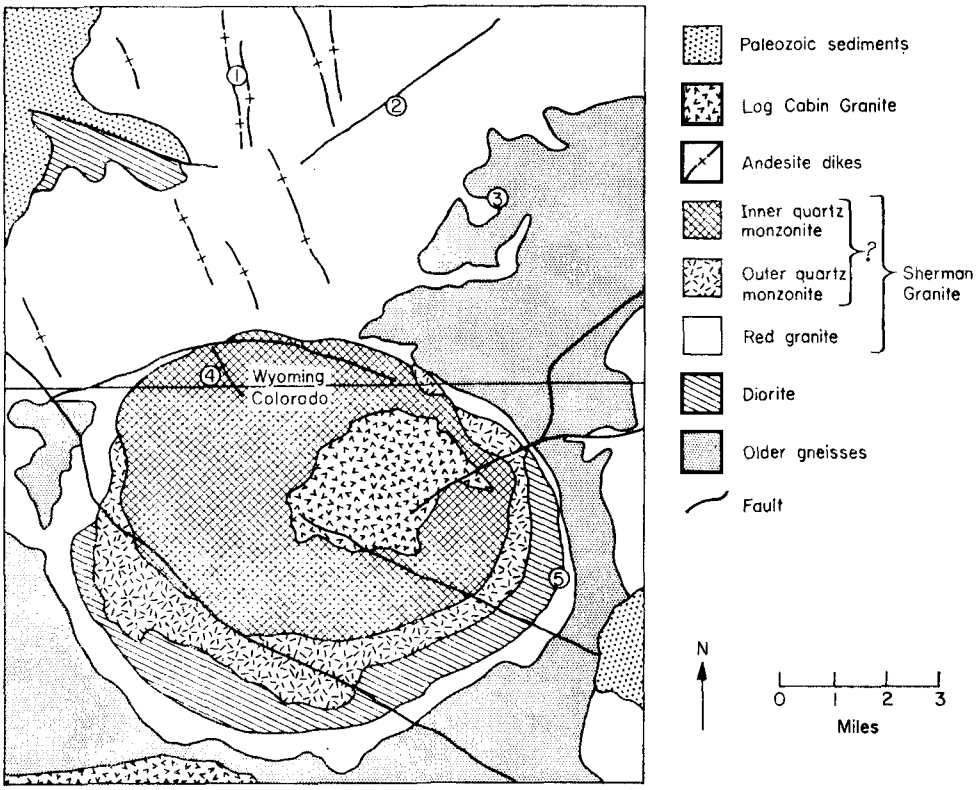


FIG. 2. Simplified geological map of the Virginia Dale area (A, Fig. 1) showing the positions of sampling sites 1–5.

Paleozoic sedimentary rocks near the Colorado–Wyoming border are essentially flat-lying, indicating that post-Pennsylvanian tilting has been minimal in that area. Southward, near Big Thompson canyon (Fig. 3), the overlying sediments strike generally north–south and dip eastward 20–25 degrees.

3. Radiogenic ages

Samples from both the red granite and quartz monzonite facies of Sherman Granite fit a precise Rb–Sr method isochron of $1{\cdot}41 \pm 0{\cdot}03$ G.Y. Concordant Rb–Sr microcline and biotite ages and K–Ar dates indicate that the Sherman has not been thermally disturbed since emplacement (Peterman *et al.* 1967). Andesite dykes are geologically older than granite of the Log Cabin batholith, which is dated at 1·42 G.Y. by Rb–Sr methods, and themselves give a K–Ar age of 1·42 G.Y. (Ferris & Krueger 1964). Therefore all the rocks sampled appear to have been emplaced within a relatively short time span, possibly 40 m.y. (Peterman *et al.* 1967).

4. Palaeomagnetic results

4.1 *Sherman Granite.* Reliable magnetic directions were obtained from some samples of the red granite facies of Sherman Granite. Polished-section studies under oil at magnifications up to 1000 reveal that the rock has been affected by high-temperature oxidation. The results of the oxidation are: (1) the alteration of mag-

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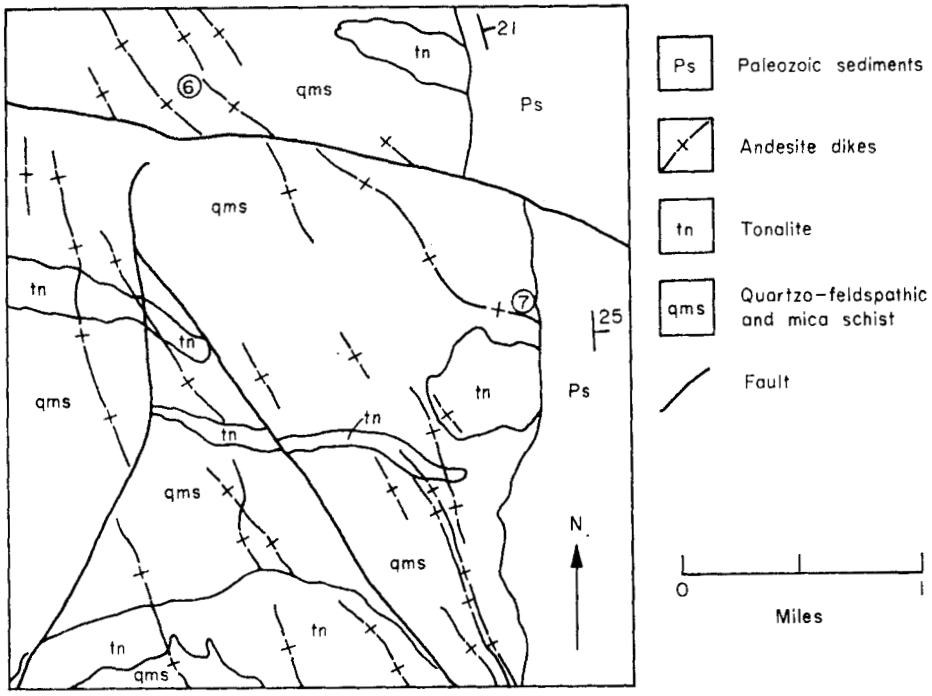


FIG. 3. Simplified geological map of the Big Thompson canyon area (B Fig. 1) showing the position of sampling sites 6–7. (After Nutalaya 1966).

netite to hematite; (2) the alteration of ilmenite to metailmenite and in some cases to pseudobrookite; (3) development of hematite lenses parallel to the cleavage direction in biotite grains; and (4) dissemination of hematite pigment, particularly along fractures and at centres within feldspar grains. The last process is responsible for the colour of the rock. Oxidation of magnetite grains occurs from the periphery inward and proceeds along the octahedral (111) crystallographic planes. In the reddest rocks, which provide the most stable directions, the oxidational effects are the most pronounced.

The relatively high P_{O_2} necessary to cause these changes was probably supplied by volatiles, largely H_2O, concentrated during the end stages of crystallization. The presence of pseudobrookite, which is stable only above 600° C. Lindsley (1966), indicates that the oxidation occurred at rather elevated temperatures. Other field evidence supports this interpretation.

Magnetic directions for five samples after a.c. demagnetization in peak fields up to 350 Oe are plotted in Fig. 4. Several other samples are rejected because they appeared to be affected by lightning or were unstable upon demagnetization.

Samples from the relatively younger quartz monzonite facies of Sherman Granite in the inner zone generally are magnetically unstable. However, two of the redder specimens give directions, after a.c. demagnetization in peak fields up to 350 Oe, parallel to those of the red granite facies, as shown in Fig. 4.

4.2 *Basic inclusions.* Gabbroic inclusions are fairly common in red granite. They are often too coarse-grained to be stable, but one stable sample gave a direction similar to red granite (Fig. 4).

The quartz monzonite facies of the Sherman Granite in the outer zone of the ring-dyke contains angular xenoliths of fine-grained diorite from which reliable directions can be obtained (Fig. 4). Small opaque grains in these rocks are magnetite and ilmenite.

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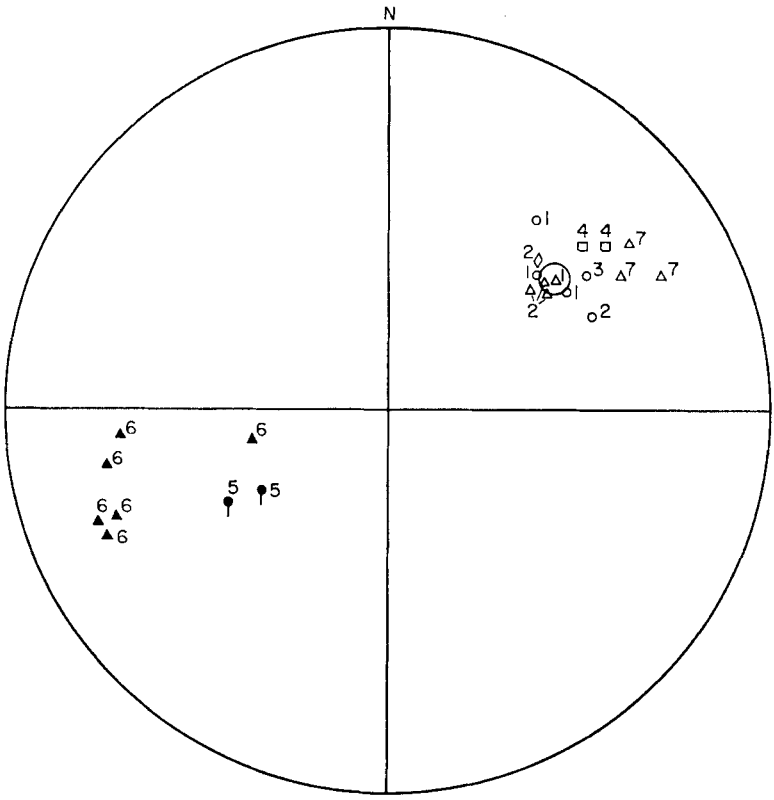


FIG. 4. Directions of magnetization of Front Range Pre-Cambrian samples in equal-area projection. Circles indicate red Sherman Granite, squares inner quartz monzonite facies of Sherman Granite, triangles andesite dykes, diamond gabbro inclusions in red Sherman Granite, and circles with tails diorite inclusions in outer quartz monzonite facies. Numbers indicate sampling sites. Black symbols are used for directions on the lower hemisphere and open symbols for directions on the upper hemisphere. The large open circle represents the mean direction of sites 1–5.

4.3 *Andesite dykes.* Andesite dykes have proven to be the most reliable of all rocks in the area. The dykes, 2 to 25 ft wide and up to 5 miles in length, are composed principally of quartz, microcline, andesine, biotite and amphibole.

The opaque mineralogy is quite varied. Many dykes contain small to large irregular crystals of titanomagnetite and ilmenite. Commonly, initial high-temperature oxidation has produced relatively coarse ilmenite lamellae in the titanomagnetite grains, which may later by oxidized to metailmenite (see Larson *et al.* in press). However, others contain mostly magnetite which is altered to hematite about grain edges, and some contain a little pyrite or chromite (?). Stability is greater in chilled dyke margins and in those dykes in which there is abundant development of ilmenite lamellae in the titanomagnetite grains. There are no observable differences in opaque mineralogy between dykes magnetized up to the north-east and those magnetized antiparallel to that direction. This observation and the great variety of opaque mineralogy associated with each polarity seem to be most in accord with the idea of field reversals.

Palaeomagnetic directions after demagnetization in peak fields up to 400 Oe are plotted in Fig. 4 for sites 1 and 2. Directions in the dykes closely parallel those determined for red Sherman Granite and the xenoliths.

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Magnetic directions from dykes at the two most southerly sites (6 and 7) at Big Thompson Canyon are stable and well-grouped but are displaced about 22 degrees from the mean of the dykes (and red granite and xenoliths) discussed above. The mean direction at site 6 is antiparallel, or nearly so, to that at site 7. The angular discrepancy could be the result of secular variation. However, sites 6 and 7 lie close to the range front, along which paleozoic sedimentary beds are tilted about 25 degrees to the east. If one applies a structural correction, the mean direction essentially corresponds to that of the varied rock types to the north.

5. Mean direction and palaeomagnetic pole position

A mean direction was calculated from sites 1–5 (14 samples), which have undergone little apparent structural rotation; it has declination 52° and inclination −44°. The 95 per cent confidence cone about the mean has a half angle of 4·3° and $k = 86{\cdot}4$. The palaeomagnetic pole calculated from the mean is at 8° N, 29° E; $dp = 3{\cdot}6°$ and $dm = 5{\cdot}5°$ (see Irving 1964). This result is compared with other pre-Cambrian rocks in North America of about the same age in Fig. 5.

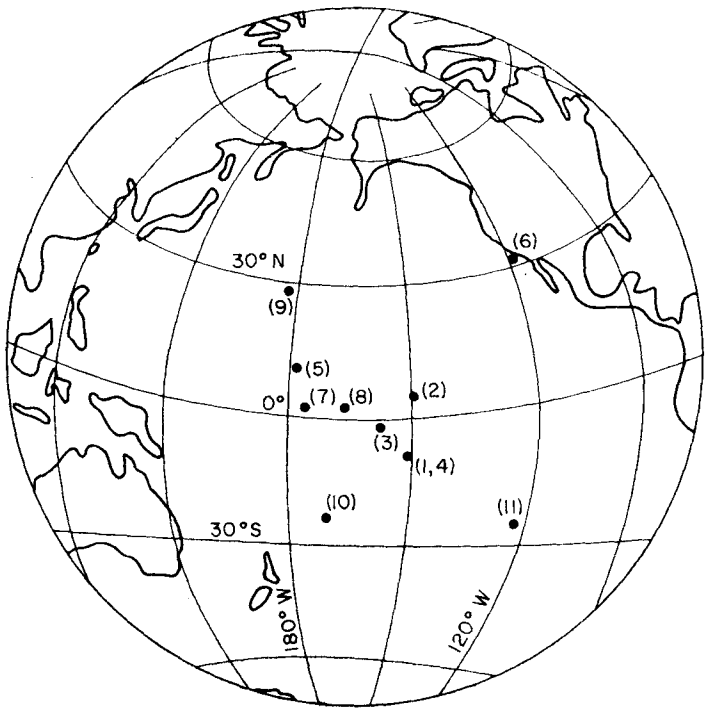


FIG. 5. Location of Pre-Cambrian geomagnetic pole position from various studies. (1) This report (south magnetic pole), rocks 1·41 G.Y. old. (2) Southeast Missouri felsite and diabase, 1·30–1·43 G.Y. (Hays & Scharon 1966). (3) Grand Canyon series combined (Irving 1964). (4) Belt Series of Montana and British Columbia combined (Irving 1964). Its age is at least 1·1 G.Y. (Runcorn 1964) but younger than 1·6 G.Y. (Giletti 1966). (5) Upper Keweenawan rocks (Irving 1964), age 0·6–1·1 G.Y. (Leech *et al.* 1963). (6) Abitibi diabase dykes, Canada, 1·23 G.Y. (7) Mackenzie dykes, 1·32 G.Y. (8) Sudbury dykes, 1·29 G.Y. (Fahrig *et al.* 1965). (9) Sierra Ancha diabase, Arizona, and Endionsill, Minnesota, 1·1 G.Y. (Helsley 1965). (10) Uinta Mountain Group (Runcorn 1964). (11) Apache Group, Arizona, age greater than 1·1 G.Y. (Runcorn 1964).

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6. *Discussion*

Gough *et al.* (1964) have suggested that the angular dispersion in Pre-Cambrian rocks in Africa is not significantly different from that of the present field and, therefore, that secular variation was quite similar. However, although the number of samples is not large, the close similarlity of palaeomagnetic directions between red Sherman Granite, diorite xenoliths, and andesite dykes tentatively suggests that the field changed little in position over a considerable length of time. This possibility is being investigated. Quartz monzonite containing diorite xenoliths is younger than the red granite but older than andesite dykes; yet its magnetic directions are opposite to those in the red granite and dykes. Apparently the field changed once in sense and then back again. The similarity in directions, therefore, seems real, and is probably not due to reheating by the dykes or to later reheating of both.

Moreover, if structural complications are adjusted, directions from dykes at Big Thompson Canyon closely align with the mean magnetic directions of the rocks to the north, apparently indicating that the field direction remained unchanged during that reversal also.

A study of possible use of palaeomagnetism in determining the amount of tectonic rotation of the Front Range is continuing.

Table 1

Relative age relations of northern Front Range rocks (younger upward) and their magnetic polarities. U denotes direction on upper hemisphere and D direction on lower hemisphere

Rock type		Polarity
Andesite dykes		*U, D*
Inner quartz monzonite	age relation	*U*
Outer quartz monzonite (inclusions)	not certain	*D*
Red granite and inclusions		*U*

Presently, there are indications of at least two, and probably four, field reversals in the northern Front Range. Relative age relations of the rocks and their polarities are shown in Table 1. The multitude of reversals in rocks of nearly the same age would seem to indicate that the rate of reversal of the Earth's field was quite rapid in the Pre-Cambrian, 1·4 G.Y. ago, perhaps approaching that of the last few million years (Cox *et al.* 1967). The large degree of anti-parallelism between upward and downward polarities seems to strongly support the idea of a dipolar field at that time.

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